# Covariance and Correlation

## Probability Theory: Foundation for Data Science

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# Expectation, Variance, Covariance, and Correlation

At the end of this module, students should be able to

- Compute the mean, variance, and standard deviation of a function of a random variable (i.e. $g(X)$).
- Explain the concept of jointly distributed random variables, for two random variables $X$ and $Y$.
- **Define, compute, and interpret the covariance between two random variables $X$ and $Y$.**
- **Define, compute, and interpret the correlation between two random variables $X$ and $Y$.**

Example: An insurance agency services customers who have both a homeowner's policy and an automobile policy. For each type of policy, a deductible amount must be specified. For an automobile policy, the choices are \$100 or \$250 and for the homeowner's policy, the choices are \$0, \$100, or \$200.

Suppose the **joint probability table** is given by the insurance company as follows:

| | | y (home) | | |
|---|---|---|---|---|
| | | 0 | 100 | 200 |
| x (auto) | 100 | .20 | .10 | .20 |
| | 250 | .05 | .15 | .30 |

When two random variables, $X$ and $Y$, are not independent, it is frequently of interest to assess how strongly they are related to each other.

Definition: The **covariance** between two rv's, $X$ and $Y$, is defined as:

Definition: Covariance of $X$ and $Y$ is given by
$Cov(X, Y) = E[(X - E(X))(Y - E(Y))]$

To calculate covariance:

$$Cov(X, Y) = \begin{cases} \sum_x \sum_y (x - \mu_X)(y - \mu_Y) P(X = x, Y = y) \\ \\ \int_{-\infty}^{\infty} \int_{-\infty}^{\infty} (x - \mu_X)(y - \mu_Y) \underbrace{f(x, y)}_{\text{Joint-density function}} \, dx \, dy \end{cases}$$

The covariance depends on both the set of possible pairs and the probabilities for those pairs.

$$Cov(X, Y) = E[(X - E(X))(Y - E(Y))]$$

- If both variables tend to deviate in the same direction (both go above their means or below their means at the same time), then the covariance will be positive.
- If the opposite is true, the covariance will be negative.
- If $X$ and $Y$ are not strongly (linearly) related, the covariance will be near 0.

Covariance example calculation:

| | | y (home) | | |
|---|---|---|---|---|
| | | 0 | 100 | 200 |
| x (auto) | 100 | .20 | .10 | .20 |
| | 250 | .05 | .15 | .30 |

Computational formula for covariance:
$Cov(X, Y) = E(XY) - E(X)E(Y)$

What if $X$ and $Y$ are independent?

Useful formulas for random variables $X$ and $Y$ and real numbers $a$ and $b$:

- $E(aX + bY) = aE(X) + bE(Y)$
- $V(aX + bY) = a^2V(X) + b^2V(Y) + 2abCov(X, Y)$

Definition: The **correlation coefficient** of $X$ and $Y$, denoted by $Cor(X, Y)$ or just $\rho_{X,Y}$, is defined by

It represents a "scaled" covariance. The correlation is always between -1 and 1.

Two special cases:

- What if $X$ and $Y$ are independent?

- What if $Y = aX + b$?

| | | y (home) | | |
|---|---|---|---|---|
| | | 0 | 100 | 200 |
| x (auto) | 100 | .20 | .10 | .20 |
| | 250 | .05 | .15 | .30 |

Find $\rho_{X,Y}$